

Jack Levitt

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EDUCATION

Princeton University

Expected May 2029

Bachelor of Science in Engineering in Mechanical and Aerospace Engineering

Cumulative GPA: 3.85/4.0

RELEVANT EXPERIENCE

Reworld Mechanical/Operations Engineering Intern

May 2026 - August 2026

- Performed and authored a plant-wide structural analysis across 50+ mechanical systems, evaluating load paths, support conditions, and equipment integrity to identify areas requiring engineering review and improve maintenance planning.
- Developed and revised P&IDs and CAD models for plant equipment and utility systems, producing accurate engineering drawings that supported communications between operations, maintenance, and external vendors.
- Developed technical engineering documentation detailing component mechanics, failure modes, and maintenance principles of critical plant equipment to improve operator understanding of system functionality.

Princeton Rocketry Recovery Systems Lead

September 2025 - Present

- Led recovery systems engineering for 30,000-ft high-powered rockets, developing parachute deployment mechanisms and conducting ground testing to validate mission-critical flight hardware.
- Performed CFD simulations of rocket geometries to evaluate airflow behavior, aerodynamic stability, and performance impacts across flight regimes.
- Designed, fabricated, and validated optimized bulkhead structures using CAD and manufacturing processes, improving structural performance while optimizing subsystem mass.

Orbis-Metallicum LLC Manufacturing Systems Engineering Intern

December 2025 - Present

- Built and optimized a global manufacturing database of EV gigacasting, integrating casting architecture, machine tonnage, and component-level design specifications across 12 OEMs and 40+ facilities, comprising 2,000+ data points.
- Evaluated graphene and graphene-doped aluminum/copper composites as next-generation electrical conductors, benchmarking theoretical thermal and electrical conductivity gains against incumbent materials to assess feasibility for weight-sensitive applications in aerospace and high-power electronics.
- Developed first-principles models of automotive battery chemistries, analyzing material composition, manufacturing processes, and electrochemical performance tradeoffs to evaluate future technology adoption.

Princeton Student Ventures Director of Events

September 2025 - Present

- Cultivated relationships with technical founders, venture capital investors, and industry leaders to accelerate the commercialization of emerging engineering and deep-tech innovations.
- Analyzed emerging deep-tech industries and startup ecosystems to identify founders and technologies driving innovation across advanced manufacturing, robotics, aerospace, and energy.
- Evaluated the technical feasibility and commercialization potential of emerging technologies by examining product development strategies, engineering challenges, and pathways from innovation to market adoption.

RESEARCH EXPERIENCE

Princeton University Rabitz Lab – Undergraduate Research Assistant

August 2026 - Present

- Collected and analyzed cryogenic infrared spectra of biological small molecules to generate high-quality reference datasets, directly supporting development of a novel microscopy instrument designed to map the spatial distribution of molecules within cells in their native state.
- Performed spectroscopy experiments under cryogenic conditions, applying data processing and spectral analysis techniques to characterize biological molecules while ensuring reproducible, high-fidelity experimental results.
- Collaborated with faculty and research mentors to interpret infrared spectroscopy data and validate reference spectra, accelerating efforts to develop new methods for studying cellular metabolism and molecular organization.

TECHNICAL SKILLS

Engineering Software: MATLAB, Python, C++, SolidWorks, PTC Creo, Autodesk Fusion 360, AutoCAD, OpenRocket, LaTeX

Analysis & Design: Computational Fluid Dynamics (CFD), CAD Modeling, Structural Analysis, Manufacturing Design, Systems Engineering

Other: Microsoft Suite, Spanish (Intermediate), NYS Seal of Biliteracy